

HLT Diphoton Trigger Efficiency — 2024 Low-Mass

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CMS low-mass $H \rightarrow \gamma\gamma$

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Results repo: [HLT-diphoton-efficiency-2024-results](#) · Interactive explorer: [GitHub Pages](#)

Overview

- **Measurement:** tag-and-probe (TnP) efficiency of the *two legs* of the CMS diphoton high-level-trigger path, on **2024 data**, using $Z \rightarrow ee$ electrons as photon probes.
- **Why reweight:** $Z \rightarrow ee$ probe photons have different $(R_9, |\eta|)$ distributions than $H \rightarrow \gamma\gamma$ signal photons, so each probe is reweighted in $(R_9, |\eta|)$ to resemble the $H \rightarrow \gamma\gamma$ population *before* the efficiency fit.
- **2024 scope** — full 2024 data (EGamma{0,1}, eras C–I), crossed along two axes:
 - η - R_9 reweighting derived noPresel / withPresel (without / with the photon preselection);
 - trigger selection: standard, or additionally requiring the full HLT path to have fired.
- Two legs (*seeded* / *unseeded*), five η/R_9 categories, probe- E_T bins.
- **Deliverables:** public results repo, a production/verification PDF, and an interactive cross-year explorer (2022/2023/2024).

Shown here: the η - R_9 -reweighted result (reweighting derived without preselection) — standard, then full HLT path fired — plus comparisons to 2023 and to the previously-derived 2022 efficiencies.

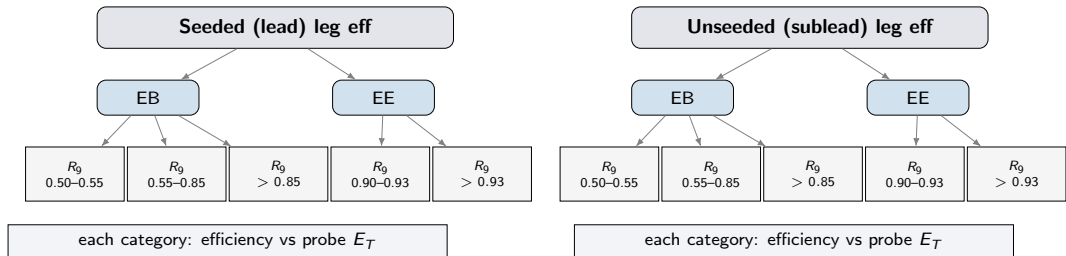
Tag-and-Probe technique

The “Tag-and-Probe” technique measures the trigger efficiency ($Z \rightarrow ee$ target topology), using the EGM POG `egm-tnp-analysis` framework:

- **Probe selection:** seeded leg $|\eta| \leq 2.5$, $p_T > 30$ GeV; unseeded leg $|\eta| \leq 2.5$, $p_T > 18$ GeV.
- **Signal model:** **bwDSCB** (Breit–Wigner $\otimes$ double-sided Crystal Ball).
- η – R_9 reweighting applied (Section above) before extracting the photon efficiency.

$$\varepsilon = \frac{N_{\text{pass}}^{\text{sig}}}{N_{\text{pass}}^{\text{sig}} + N_{\text{fail}}^{\text{sig}}}$$

Categories: $\text{leg} \times \eta/R_9 \times \text{probe } E_T$



5 η/R_9 categories $\times$ 2 legs = 10 measurement categories, each fit in probe- E_T bins: seeded

{35–37, 37–40, 40–45, 45–50, 50–60, 60–70, 70–90, >90} GeV (8 bins); unseeded

{21–25, 25–28, 28–31, 31–35, 35–40, 40–45, 45–50, 50–60, 60–70, 70–90, >90} GeV (11 bins). Crossed with the 4 reweighting/selection variants: $5 \times 2 \times 4 = 40$ (category $\times$ leg $\times$ variant) combinations.

Dataset — 2024

Data = TnP probe photons; **MC** = gluon-fusion $H \rightarrow \gamma\gamma$ ($m_H = 90$) signal, used only to build the η - R_9 weight map.

	Data	MC (for η-R_9 weights)
sample probes	EGamma0 + EGamma1, Run2024 C-I ~ 160 M	GluGlu $H \rightarrow \gamma\gamma$, $m_H = 90$ ~ 2.4 M

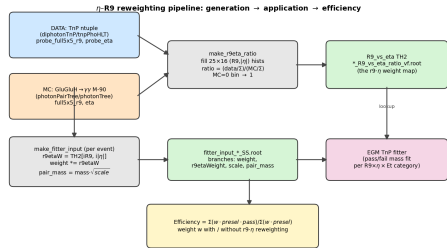
- **Data:** full 2024, both EGamma primary datasets, eras **C through I**; integrated luminosity $\approx 109 \text{ fb}^{-1}$.
- **MC (weights only):** gluon-fusion $H \rightarrow \gamma\gamma$ signal at $m_H = 90$ GeV — gives the $H \rightarrow \gamma\gamma$ photon ($R_9, |\eta|$) shape the data probes are reweighted to.
- Weight-map binning: $25 R_9 \times 16 |\eta|$ bins.

η - R_9 reweighting — what and why

- Probe photons from $Z \rightarrow ee$ do not have the same $(R_9, |\eta|)$ shape as $H \rightarrow \gamma\gamma$ signal photons.
- Build an MC/data ratio map in $(R_9, |\eta|)$ (density-normalised) — target over source — and multiply each probe by its cell value:

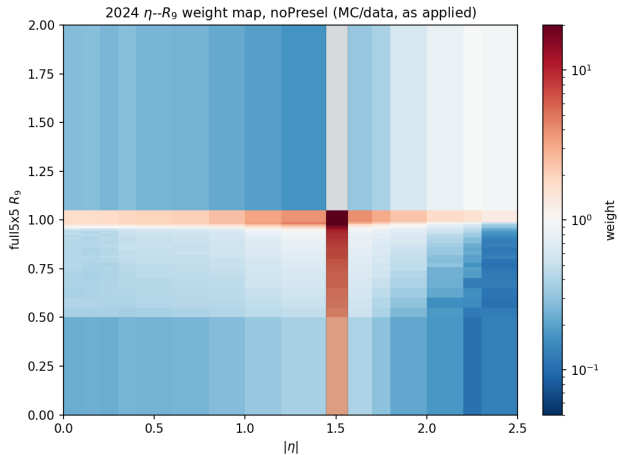
$$w_{ab} = \frac{N_{ab}^M / \sum N^M}{N_{ab}^D / \sum N^D}, \quad w_{ab} = 1 \text{ if } N_{ab}^M = 0.$$

- It is a per-event **multiplicative weight**, *not* a cut — it reweights the probe population, applied to both pass and fail.
- Two variants: noPresel (map from all probes) vs withPresel (map after the offline preselection).



Generate map → apply per event → enters the TnP fit as a weight.

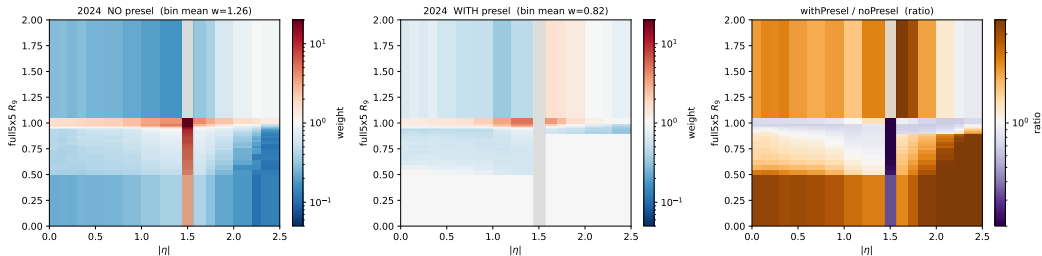
η - R_9 2-D weight map (2024, noPresel)



- 2-D histogram of the MC/data ratio in $(R_9, |\eta|)$.
- Log colour scale, pivot at $w = 1$ (white).
- Blue ($w < 1$): low- R_9 / endcap probes are down-weighted (cell mean 0.68 below $R_9 = 0.8$; 0.32 where both hold).
- Red ($w > 1$): high- R_9 probes are up-weighted (cell mean 1.91 above $R_9 = 0.9$) — $H \rightarrow \gamma\gamma$ photons are more high- R_9 than $Z \rightarrow ee$ probes.
- Red band at $R_9 \approx 0.95$ – 1.0 and the grey EB–EE gap column (1.44 – 1.57) are common to all years; same 25×16 binning as the marginals.

η - R_9 weight map: noPresel vs withPresel (2024)

2024 η - R_9 weight map, MC/data as applied: noPresel vs withPresel (log colour, pivot 1.0)



Same $(R_9, |\eta|)$ map, built without vs with the settings preselection applied at fill time; right panel is the ratio. `withPresel` pulls the map's mean cell weight toward 1 ($1.26 \rightarrow 0.82$; the map is built on a cleaner, preselected probe population) — the strong low- R_9 /high- $|\eta|$ down-weighting is markedly milder ($0.32 \rightarrow 1.00$).

Preselection — all cut values

Same selection applied in every fit bin (both reweighting variants use identical cuts).

Tag (common)

variable	cut
p_T	$> 35 \text{ GeV}$
$ \eta_{\text{SC}} $	< 2.5
ID MVA	> -0.6

Probe — common to both regions

variable	cut
p_T (seeded / unseeded)	$> 30 / > 18 \text{ GeV}$
ID MVA	> -0.9
H/E	< 0.08
photon iso (ρ -corr)	< 3.0
track iso (cone 0.3)	< 6.0

Probe — region-dependent

variable	EB	EE
$ \eta_{\text{SC}} $	< 1.4442	$1.566\text{--}2.5$
R_9 floor	> 0.5	≥ 0.90
$\sigma_{i\eta i\eta}$	< 0.011	< 0.032

$H \rightarrow \gamma\gamma$ **isolation** (OR of):

$R_9 > 0.8$ OR $\text{chg.iso} \cdot p_T < 20$ OR
 $\text{chg.iso} < 0.3$

R_9 **categories** (per-bin):

EB: $0.50\text{--}0.55$, $0.55\text{--}0.85$, > 0.85

EE: $0.90\text{--}0.93$, > 0.93

Unseeded leg additionally requires the tag to have fired the seeded leg. “Full HLT path fired” variant additionally requires the full path to have fired. Numerator = probe passes the leg.

TnP fitter — how the fits are run

CMS EGamma POG `egm_tnp_analysis` (CMSSW_10_6_8). For each category $\times$ leg the driver calls `tnpEGM_fitter.py` through a fixed sequence:

```
# run_HLT_2024_all.sh <WEIGHT> <SET>   WEIGHT=noPresel|withPresel  SET=normal|fullPath
for CAT in EB_R9_0p50to0p55 EB_R9_0p55to0p85 EB_R9gt0p85 EE_R9_0p90to0p93 EE_R9gt0p93; do
  for LEG in seed unseed; do
    FITTER $CFG --flag $FLAG --checkBins      # sanity-check bin definitions
    FITTER $CFG --flag $FLAG --createBins      # build (ph_sc_abseta, ph_sc_et) bins
    FITTER $CFG --flag $FLAG --createHists     # fill pass/fail pair_mass hists (WEIGHTED)
    FITTER $CFG --flag $FLAG --doFit --bwDscb   # BWxDSCB signal + RooCMSShape bkg (nominal)
    FITTER $CFG --flag $FLAG --doFit --bwDscbExp # BWxDSCB signal + Exponential bkg (
      alternate)
    FITTER $CFG --flag $FLAG --sumUp           # write per-bin efficiency (eff, +err, -err)
  done
done
```

`--createHists` fills, per (η, E_T) bin, a *passing* and a *failing* m_{ee} histogram (pair_mass, 80 bins, [60, 120] GeV) with the per-event weight; “passing” = `probe_passSeed==1` (or `probe_passUnseed==1`).

Reweighting + preselection inside the fitter

The η - R_9 weight is looked up per event, written as a branch in the input tree, selected by the settings file, and applied by the fitter when it fills the mass histograms:

```
# (1) per-event lookup (input-tree preparation): each probe -> one weight
ix = clip(digitize(r9,          R9_EDGES ) - 1, 0, nR9 -1)  # 25 R9 bins
iy = clip(digitize(abs(eta), ETA_EDGES) - 1, 0, nEta-1)    # 16 |eta| bins
r9etaWeight = weights_2d[ix, iy]                          # stored map is data/MC
weight_{no,with}_presel = weight0 * r9etaWeight           # per event in the input tree

# (2) the settings SELECT the variant and INVERT the map (no 29 GB tree rebuild):
#      weight / r9etaWeight^2 = weight0 / r9etaWeight = weight0 * (MC/data)
sample(data_2024_noPresel, lumi=109 fb^-1,
       weight = weight_no_presel/(r9etaWeight_no_presel*r9etaWeight_no_presel))
#   withPresel -> same expression with the _with_presel branches

# (3) the fitter APPLIES it filling the pass/fail mass histograms (rootUtils.py):
cutPass = '( %s && %s ) * %s' % (cuts, flag, sample.weight)  # weighted PASS
cutFail = '( %s && %s ) * %s' % (cuts, notflag, sample.weight) # weighted FAIL
```

The *same* weight enters both pass and fail templates, so ε is reweighted consistently.

noPresel/withPresel differ only in the map the weight came from; the preselection is the cutBase applied identically in both.

Fit model — functional forms

m_{ee} fit in [60, 120] GeV per bin, pass and fail separately; $\varepsilon = N_{\text{pass}}^{\text{sig}} / (N_{\text{pass}}^{\text{sig}} + N_{\text{fail}}^{\text{sig}})$.

Each spectrum = signal + background:

Signal — Breit-Wigner (fixed Z line-shape) convolved with a double-sided Crystal Ball (RooDoubleCBShape) resolution:

$$S(m) = \text{BW}(m; M_Z, \Gamma_Z) \otimes \text{DSCB}(m; \mu, \sigma, \alpha_1, n_1, \alpha_2, n_2), \quad t \equiv \frac{m - \mu}{\sigma},$$
$$\text{DSCB}(t) \propto \begin{cases} A_1 (B_1 - t)^{-n_1}, & t < -\alpha_1 \\ e^{-t^2/2}, & -\alpha_1 \leq t \leq \alpha_2 \\ A_2 (B_2 + t)^{-n_2}, & t > \alpha_2 \end{cases} \quad A_i = \left(\frac{n_i}{|\alpha_i|} \right)^{n_i} e^{-\alpha_i^2/2}, \quad B_i = \frac{n_i}{|\alpha_i|} - |\alpha_i|.$$

Background — nominal **RooCMSShape** (error-function turn-on $\times$ exponential);
alternate **Exponential** (background systematic):

$$B_{\text{CMS}}(m) = \text{erfc}((\alpha_{\text{cms}} - m) \beta) e^{-\gamma(m - \text{peak})}, \quad B_{\text{exp}}(m) = e^{\alpha_{\text{bkg}} m}.$$

M_Z, Γ_Z fixed to PDG; $\mu, \sigma, \alpha_{1,2}, n_{1,2}$ float (bounds next slide). Quoted σ_+/σ_- are the asymmetric statistical fit errors on ε .

Fit parameters — variables, start values and bounds

Same initialisation for pass (P) and fail (F); values [start, low, high] taken verbatim from the 2024 settings files.

Signal — double-sided Crystal Ball

param	start	low	high
μ (mean)	0.0	-3.0	3.0
σ	1.5	0.3	5.0
α_1 (left)	1.5	0.5	5.0
n_1 (left)	2.0	1.0	10.0
α_2 (right)	1.5	0.5	5.0
n_2 (right)	2.0	1.0	10.0

Background — RooCMSShape (nominal)

param	start	low	high
α_{cms}	60	50	100
β	0.05	0.01	0.08
γ	0.1	0	1
peak	90.0 (fixed)		

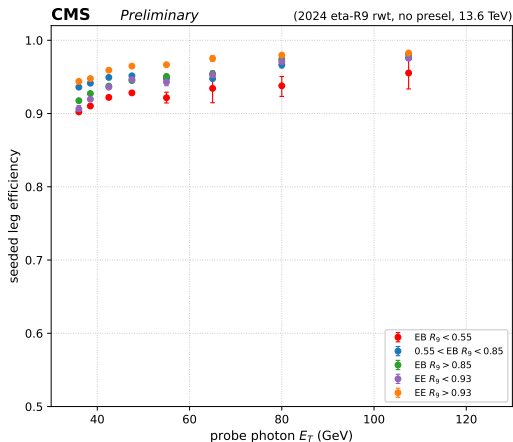
Background — Exponential (alt.)

param	start	low	high
α_{bkg}	0.0	-2.0	1.0

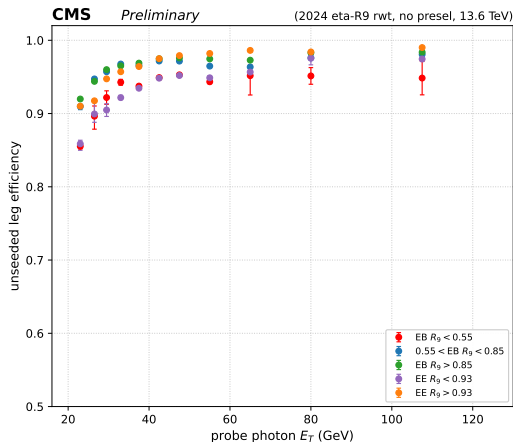
Fit variable `pair_mass`, 80 bins in [60, 120] GeV; Breit-Wigner M_Z, Γ_Z fixed to PDG values. Yields

$N_{\text{pass/fail}}^{\text{sig}}$, N^{bkg} float freely.

Results — efficiency vs E_T



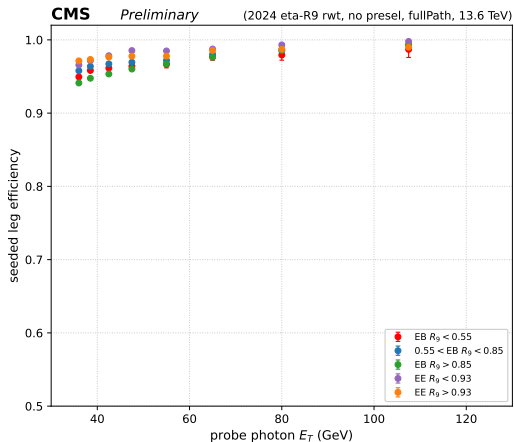
Seeded (lead) leg.



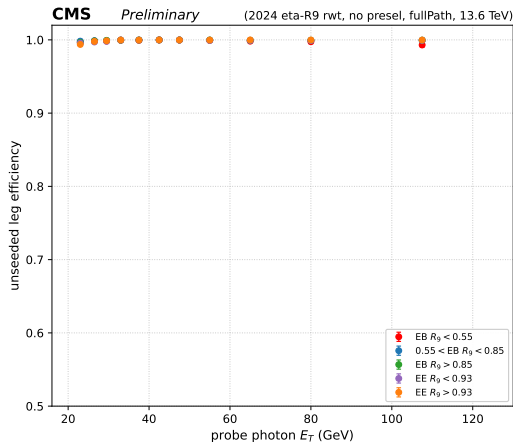
Unseeded (sublead) leg.

η - R_9 -reweighted, standard track-isolation selection; nominal bwDSCB $\otimes$ RooCMSShape fit; five η/R_9 categories overlaid.

Results — efficiency vs E_T (full HLT path fired)



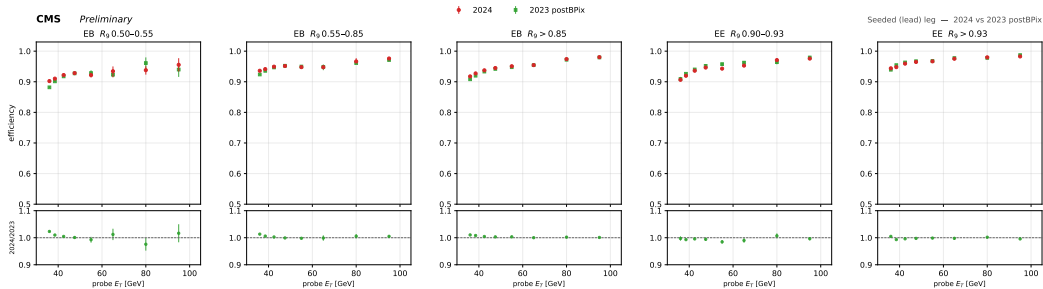
Seeded (lead) leg.



Unseeded (sublead) leg.

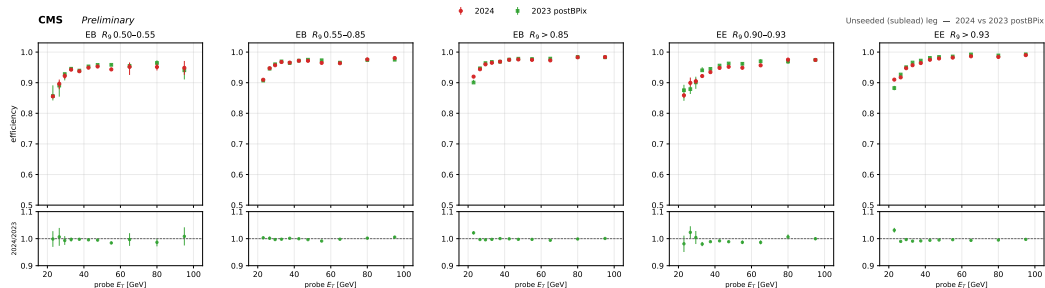
Same result, additionally requiring the full HLT path to have fired.

2024 vs 2023 — seeded leg



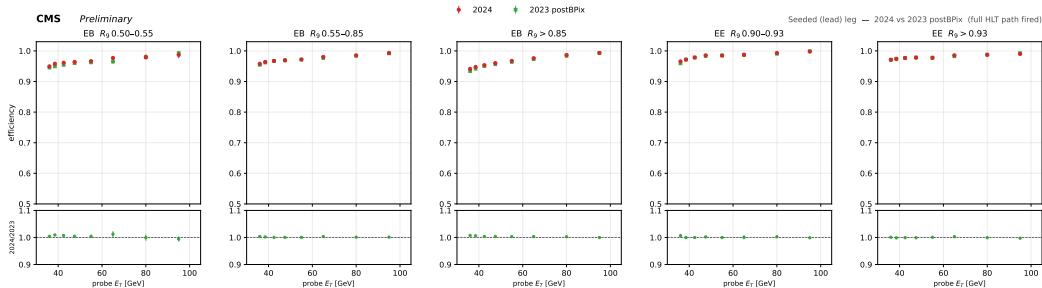
Per-category efficiency vs E_T : **2024** overlaid with **2023 postBPix**; bottom row is the ratio 2024/2023. Built from the CSV values that back the result repos. Agreement within a few percent across all five η/R_9 categories.

2024 vs 2023 — unseeded leg



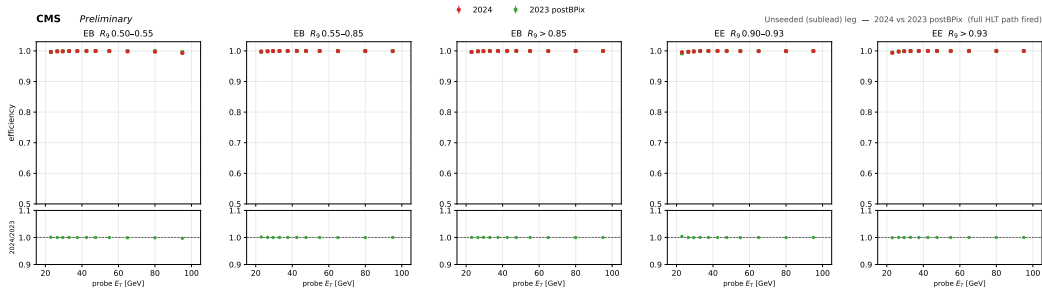
Unseeded (sublead) leg; same overlay + ratio to 2023 postBPix.

2024 vs 2023 — seeded leg (full HLT path fired)



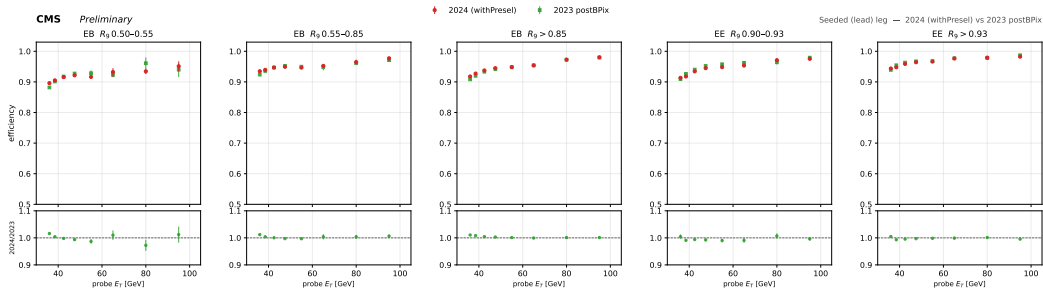
Full HLT path fired: **2024 vs 2023 postBPix**, seeded leg.

2024 vs 2023 — unseeded leg (full HLT path fired)



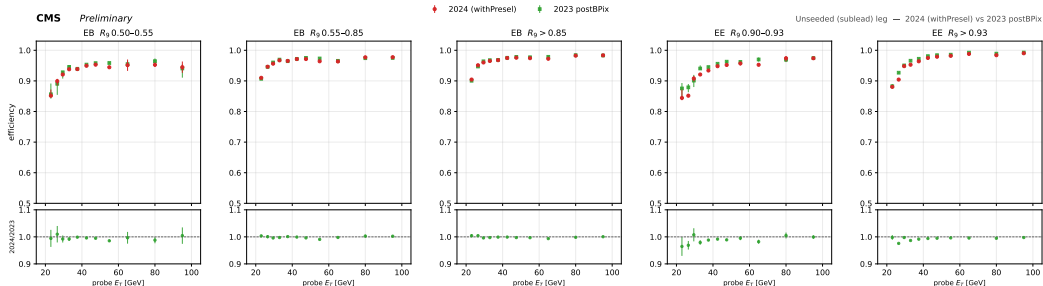
Full HLT path fired: **2024 vs 2023 postBPix**, unseeded leg.

2024 (withPresel) vs 2023 — seeded leg

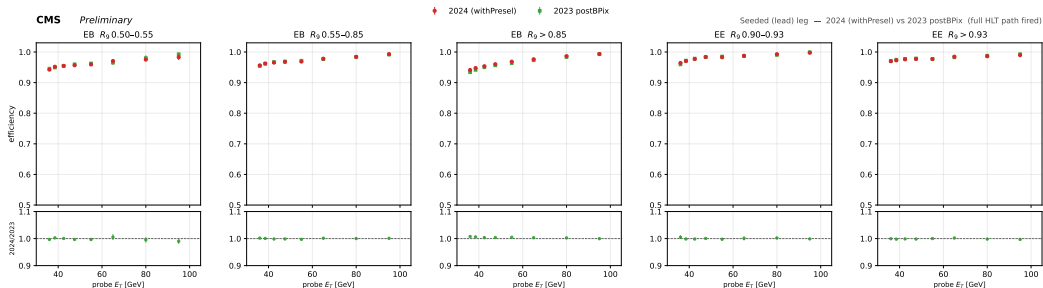


Same comparison as before, using the **withPresel** η – R_9 reweighting derivation for 2024 instead of **noPresel**.

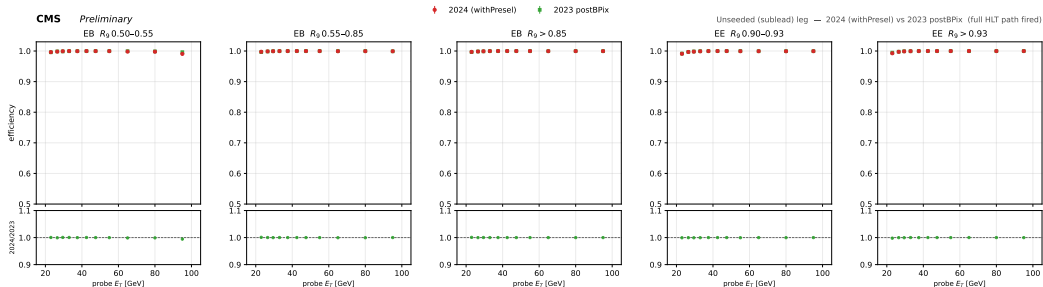
2024 (withPresel) vs 2023 — unseeded leg



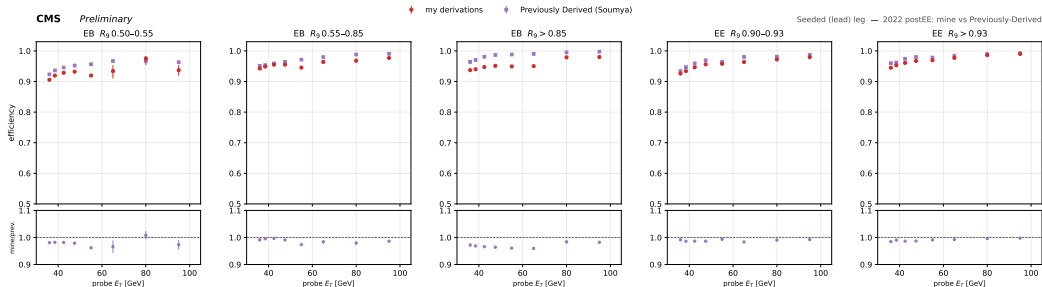
2024 (withPresel) vs 2023 — seeded leg (full HLT path fired)



2024 (withPresel) vs 2023 — unseeded leg (full HLT path fired)

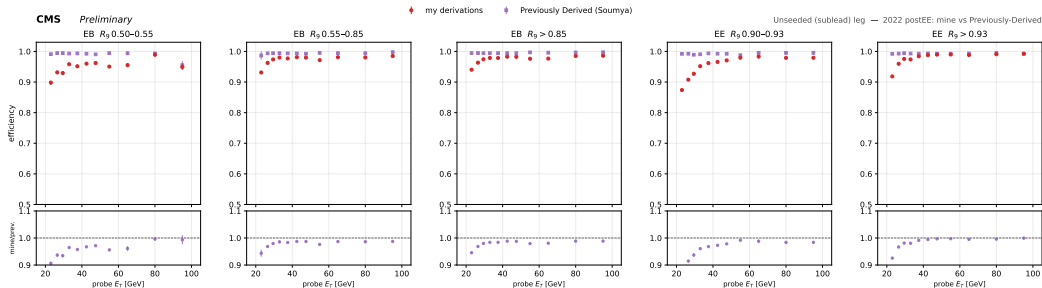


2022 — mine vs previously derived (seeded)



My derivations (bwDSCB fitter) vs the **previously-derived** 2022 postEE efficiencies (Soumya, correctionlib); ratio mine/previous below. Seeded leg.

2022 — mine vs previously derived (unseeded)



Unseeded leg. Full interactive cross-year overlay — drag/click any curve, hover a point for its mass fit
— at the [GitHub Pages explorer](#).

Summary

- First 2024 low-mass diphoton HLT leg efficiencies: tag-and-probe, η - R_9 reweighted, both legs, five η/R_9 categories, four variants (reweighting derived with/without preselection $\times$ standard / full-HLT-path).
- η - R_9 reweighting built from a 25×16 MC/data map ($H \rightarrow \gamma\gamma$ M-90 MC), applied as a per-event weight to the mass templates — not a cut.
- Signal bwDSCB $\otimes$ {RooCMSShape (nominal), Exponential (alt)}; efficiency from fitted pass/fail signal yields; asymmetric statistical errors.
- Results published; cross-year overlay in the interactive explorer.

Repo: [HLT-diphoton-efficiency-2024-results](https://github.com/santanumahata/HLT-diphoton-efficiency-2024-results) · Explorer:
santanumahata.github.io/HLT-diphoton-efficiency-2024-results

Full per-bin values + method: `HLT_efficiency_production_2023_2024.pdf`

Backup

Cut-and-count cross-check: weighted pass-fraction (no background subtraction, no fit), overlaid on the nominal fit — same categories, same E_T bins.

Backup — Production workflow

1. **Data** → **TnP trees**. CMSSW analyzer over 2024 data via CRAB ⇒ per-event tag-and-probe ntuples.



2. **Merge** + η - R_9 **reweighting**. Merge and attach the per-event η - R_9 weight ⇒ fitter-input trees (seeded / unseeded).



3. **Tag-and-probe fits**. EGM POG `egm-tnp-analysis` (CMSSW_10_6_8): fit the m_{ee} spectrum, pass and fail, in each (η, R_9, E_T) bin ⇒ per-bin efficiency.

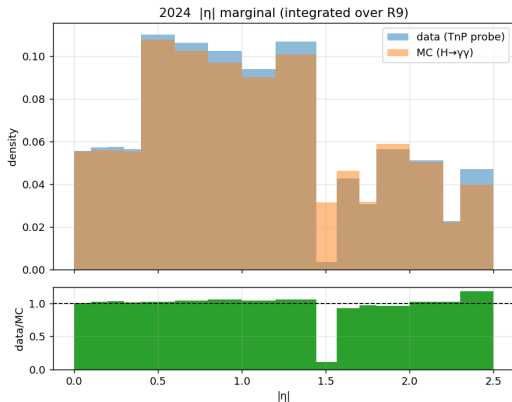
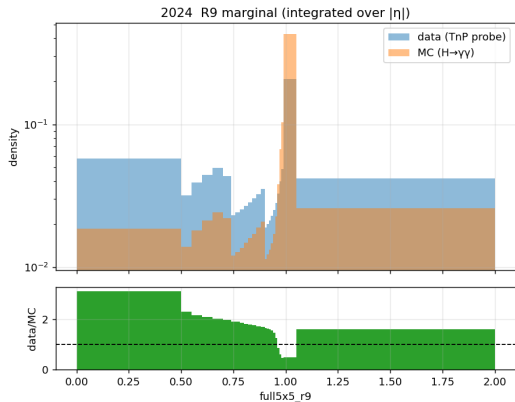


4. **Consolidation**. Collect all per-bin efficiencies into one table; build efficiency-vs- E_T plots, copy per-bin mass fits.



5. **Publish**. Push table, plots, fits to the GitHub repo; host the interactive explorer on GitHub Pages.

Backup — η - R_9 distributions that drive the weights (2024)

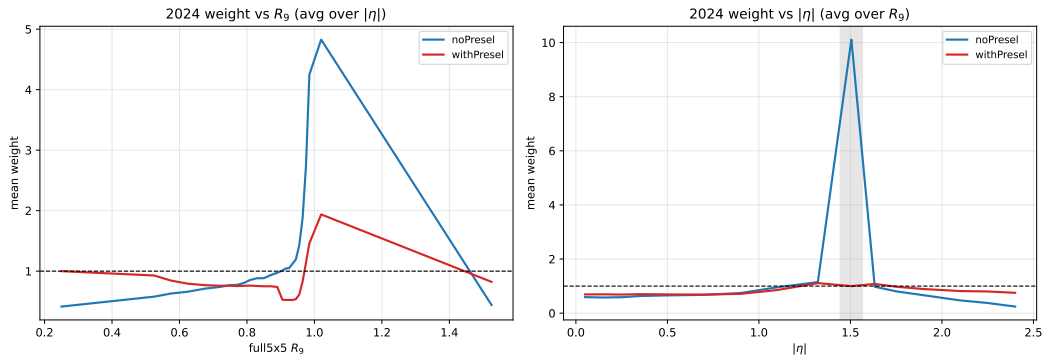


R_9 marginal (integrated over $|\eta|$), data vs MC + ratio.

$|\eta|$ marginal (integrated over R_9), data vs MC + ratio.

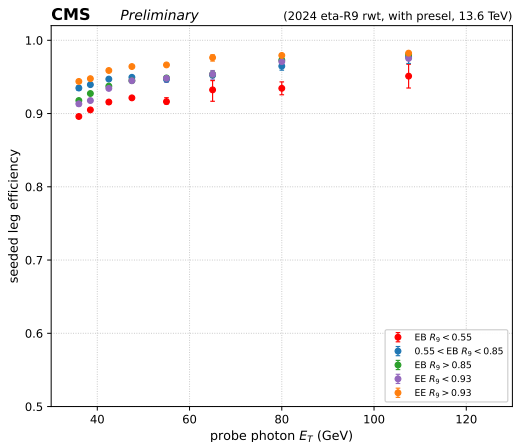
The reweighting acts most strongly at **low** R_9 and in the **endcaps**, where the $Z \rightarrow ee$ data probes and $H \rightarrow \gamma\gamma$ MC differ most; the $|\eta|$ shapes agree well after density-normalisation (ratio ≈ 1).

Backup — η - R_9 weight marginals: noPresel vs withPresel (2024)

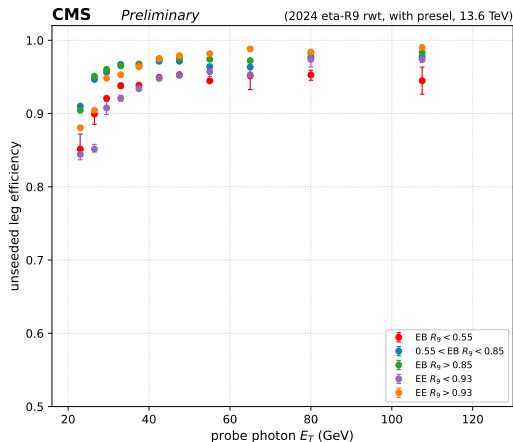


Mean weight vs R_9 (left) and vs $|\eta|$ (right), noPresel and withPresel overlaid. withPresel is flatter and closer to 1 across most of the range — the preselection removes much of the low- R_9 /high- $|\eta|$ population, where the noPresel weights are most strongly suppressed.

Backup — efficiency vs E_T , with Presel



Seeded (lead) leg.

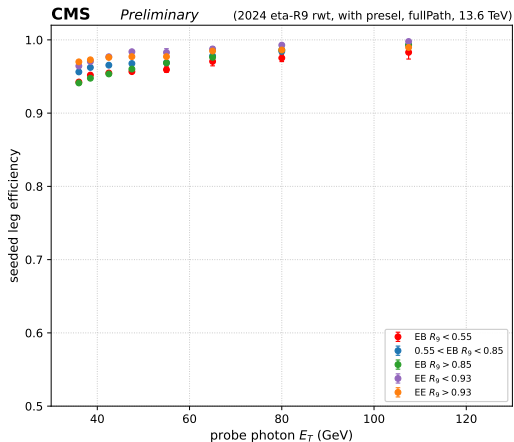


Unseeded (sublead) leg.

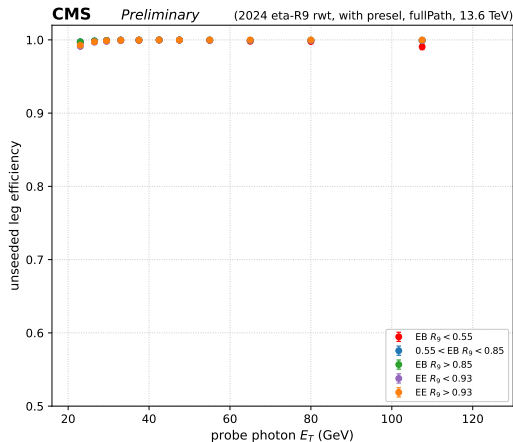
η - R_9 reweighting derived **with** preselection, standard track-isolation selection; nominal

bwDSCB $\otimes$ RooCMSShape fit; five η/R_9 categories overlaid. All efficiency bins, same

Backup — efficiency vs E_T , with Presel (full HLT path fired)



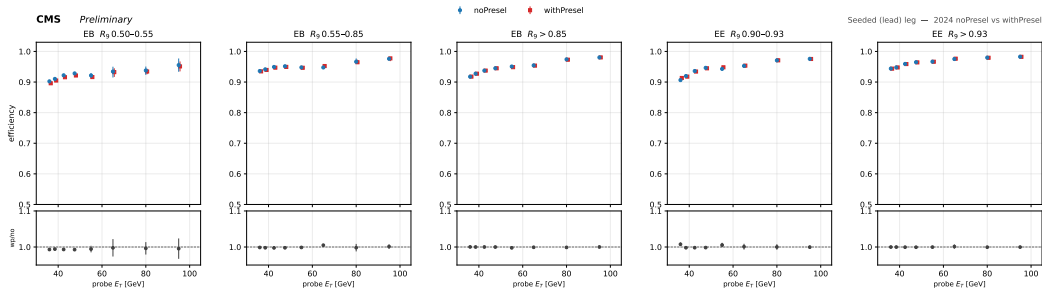
Seeded (lead) leg.



Unseeded (sublead) leg.

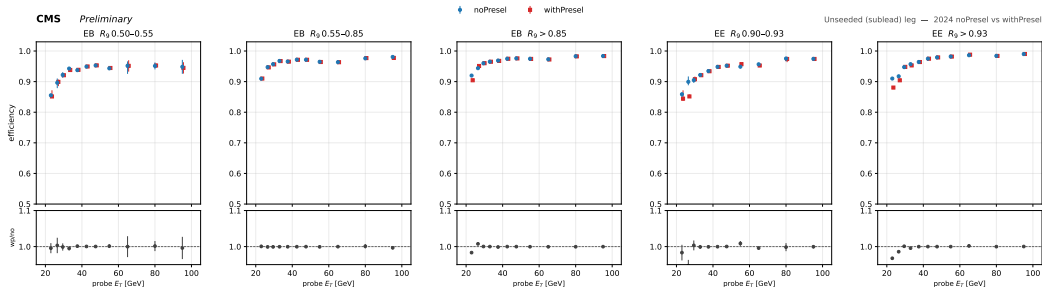
Same with Presel reweighting, additionally requiring the full HLT path to have fired.

Backup — noPresel vs withPresel efficiency ratio (seeded)

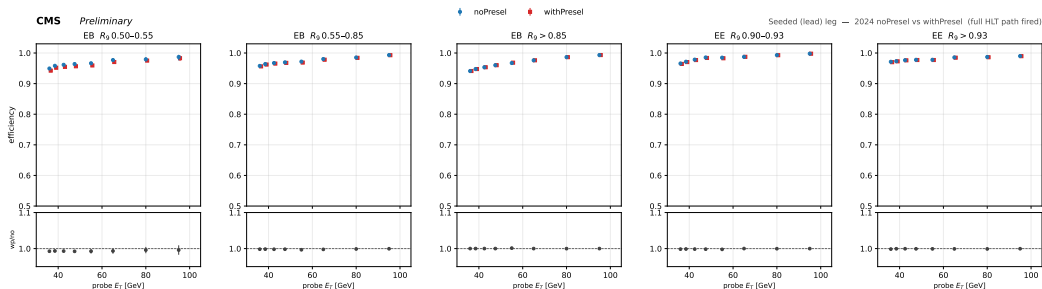


2024: same category/leg, only the reweighting-derivation choice differs. The ratio (bottom) is flat at 1 — noPresel and withPresel give essentially the same fitted efficiency, even though the underlying weight maps differ substantially (earlier slides).

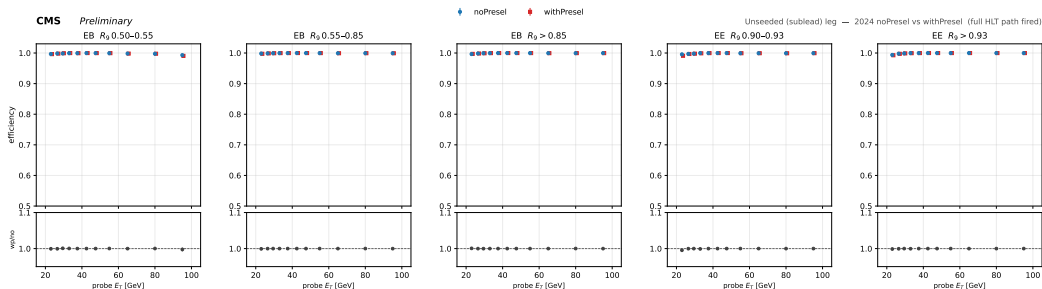
Backup — noPresel vs withPresel efficiency ratio (unseeded)



Backup — noPresel vs withPresel efficiency ratio, full HLT path fired (seeded)



Backup — noPresel vs withPresel efficiency ratio, full HLT path fired (unseeded)



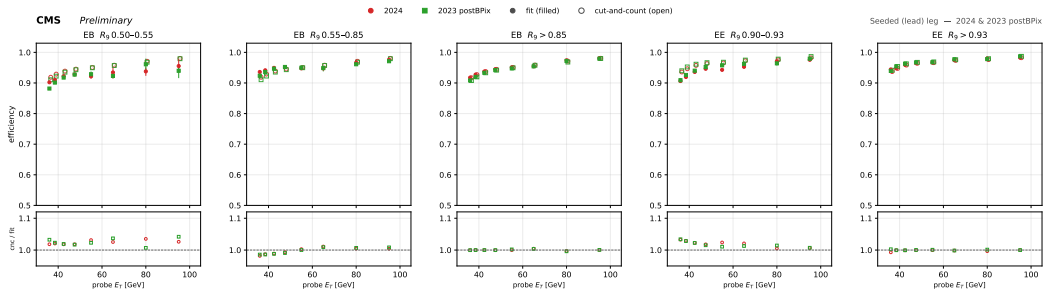
Backup — cut-and-count definition

Weighted pass-fraction on the same cutBase used by the fit, no signal/background model:

```
# per (leg, category, ET bin): same cutBase (+ passFullPath for the fullPath variant),  
# same eta-R9 weight branch as the fit, pair_mass in [60,120] GeV (the fit mass window)  
sel = eval(cutBase) & (60 < pair_mass < 120)  
eff_cnc = Sum( weight[sel & pass] ) / Sum( weight[sel] )      # pass = probe_passSeed/Unseed  
== 1
```

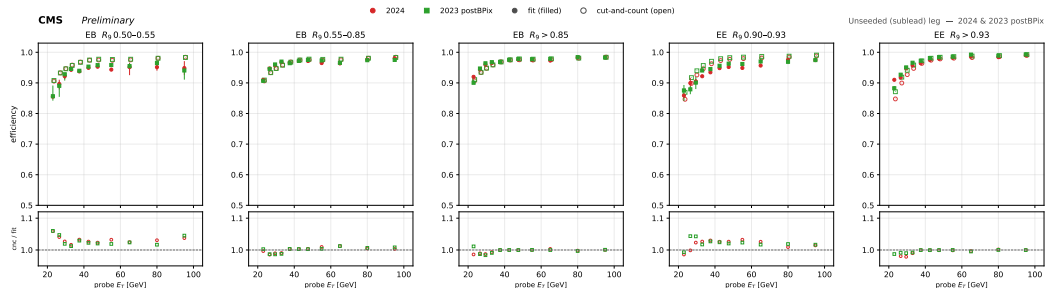
No signal/background separation — so cnc sits slightly *above* the fit (uncut background under the Z peak is not subtracted). Used only as an independent, model-free sanity check on the fitted efficiencies; **not** a competing measurement.

Backup — 2024 vs 2023 postBPix (seeded)

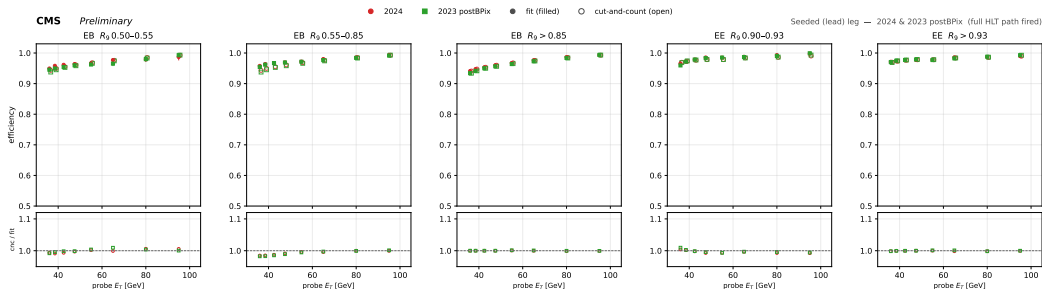


Filled = fit, open = cut-and-count; bottom row is the ratio cnc/fit.

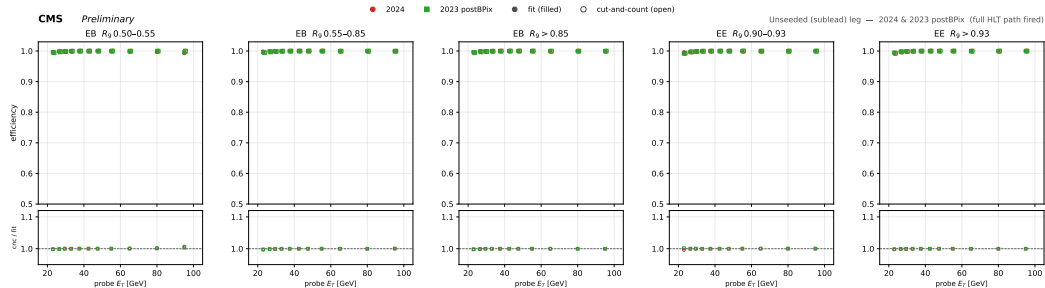
Backup — 2024 vs 2023 postBPix (unseeded)



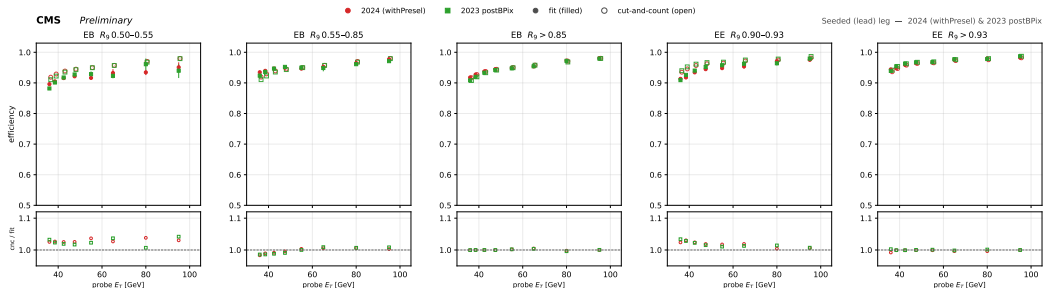
Backup — 2024 vs 2023 postBPix, full HLT path fired (seeded)



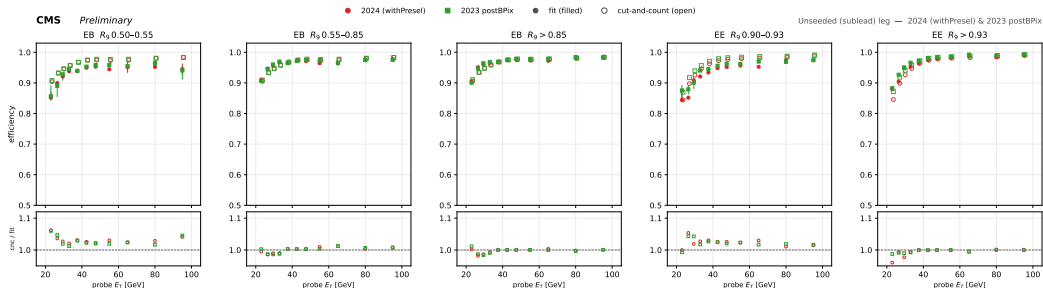
Backup — 2024 vs 2023 postBPix, full HLT path fired (unseeded)



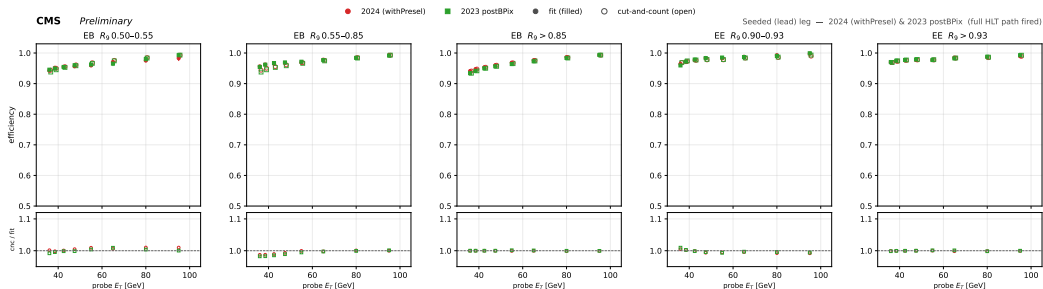
Backup — 2024 (withPresel) vs 2023 postBPix (seeded)



Backup — 2024 (withPresel) vs 2023 postBPix (unseeded)



Backup — 2024 (withPresel) vs 2023 postBPix, full HLT path fired (seeded)



Backup — 2024 (withPresel) vs 2023 postBPix, full HLT path fired (unseeded)

